

Advanced Topics in Finite Element Methods

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9th National Finite Element Developers & FEAST Users Meet
26th May 2026



Abhinav Jha

Advanced Topics in FEM, 26th May 2026



IIT Gandhinagar
MATHEMATICS

1 Convection-Diffusion-Reaction Equations

1.1 Stabilization Methods

1.2 Adaptive Mesh Refinement

1.3 Outlook & Summary



Convection-Diffusion-Reaction Equations

- Steady-state convection-diffusion-reaction equation

$$\begin{aligned} -\varepsilon \Delta u + \mathbf{b} \cdot \nabla u + cu &= f \quad \text{in } \Omega, \\ u &= u_D \quad \text{on } \Gamma_D, \\ -\varepsilon \nabla u \cdot \mathbf{n} &= u_N \quad \text{on } \Gamma_N \end{aligned}$$

- Ω – bounded domain in $\mathbb{R}^d$, $d \in \{2, 3\}$
- $\mathbf{n}$ – outward pointing unit normal
- Assume

$$\left(c(x) - \frac{1}{2} \nabla \cdot \mathbf{b}(x) \right) \geq \sigma \geq 0$$

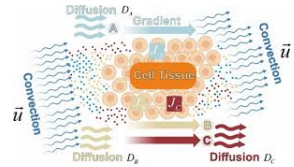
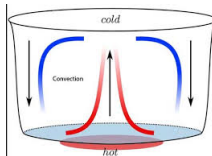
- Interested in convection-dominated regime, $\varepsilon \ll \|\mathbf{b}\|_{L^\infty(\Omega)} L$
 - L – Characteristic length of the problem



Convection-Diffusion-Reaction Equations

Convection-Diffusion-Reaction Equations

- Models Transport Phenomena



One-Dimensional Example

Convection-Diffusion-Reaction Equations

- Consider

$$-\varepsilon u'' + u' = 0 \quad \text{in } (0, 1), \quad u(0) = 0, \quad u(1) = 1.$$

- Analytical Solution & Numerical Solution

$$u(x) = \frac{\exp\left(\frac{x-1}{\varepsilon}\right) - \exp\left(\frac{-1}{\varepsilon}\right)}{1 - \exp\left(\frac{-1}{\varepsilon}\right)} \quad \text{and} \quad u^i = \frac{r^i - 1}{r^N - 1} \quad i = \{0, \dots, N\} \quad \text{with } r = \frac{2\varepsilon + h}{2\varepsilon - h}$$



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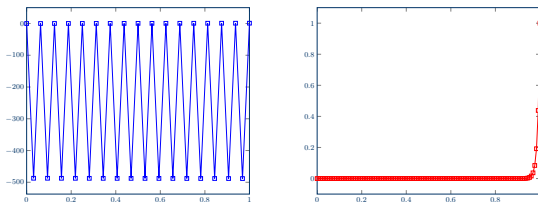


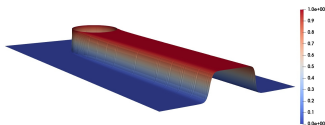
Figure 1: Numerical solution Left: $\varepsilon = 10^{-6}$, $N = 32$, Right: $\varepsilon = 10^{-3}$, $N = 128$.



Flow Around Cylinder

Convection-Diffusion-Reaction Equations

- Hemker Problem¹: $\epsilon = 10^{-4}$, $\mathbf{b} = (1, 0)^\top$, $\#\text{dof} = 33496$



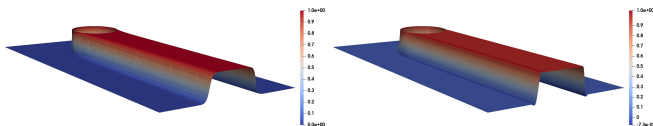
¹Hemker: *Journal of Computational and Applied Mathematics*, 76 (1-2), 277-285, 1996



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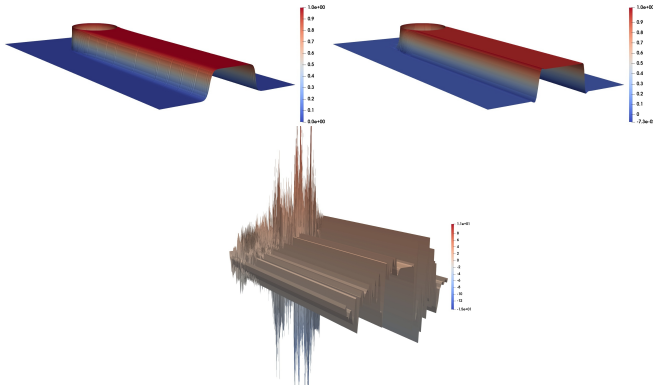
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- Ideal discretization
 - Accurate and sharp layers¹

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 - Maximum Principles

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- Ideal discretization
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 - Efficient computation of the solutions
- Alternate approach: Adaptive grids

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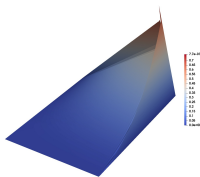
- Upwind Methods¹
 - Add artificial diffusion in the flow direction
 - Information propagated according to the wind direction
 - Removes non-physical oscillations

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- **Upwind Methods**¹

- Add artificial diffusion in the **flow direction**
- Information propagated according to the **wind direction**
- Removes non-physical oscillations
- **Limitations**
 - Excessive numerical diffusion
 - Smearing of sharp/internal layers
 - Loss of accuracy for small diffusion



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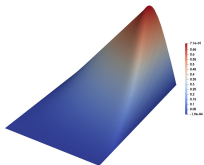
- SUPG Method²
 - Streamline Upwind Petrov-Galerkin stabilization
 - Adds stabilization only in the streamline direction
 - Preserves consistency of the formulation
 - Better accuracy compared to classical upwinding

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- Streamline Upwind Petrov-Galerkin stabilization
- Adds stabilization only in the streamline direction
- Preserves consistency of the formulation
- Better accuracy compared to classical upwinding
- Limitations
 - Oscillations may still persist near sharp layers
 - Crosswind smearing and undershoots/overshoots
 - Does not always satisfy discrete maximum principles



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Stabilization Methods

Convection-Diffusion-Reaction Equations

- AFC Schemes¹
 - Algebraic Flux Correction stabilization
 - Combines: High-order accurate scheme & Low-order monotone scheme

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 - Advantages
 - Satisfies discrete maximum principles
 - Better layer resolution compared to SUPG

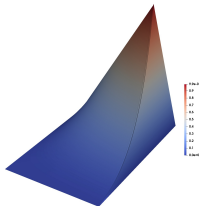
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 - Better layer resolution compared to SUPG
 - Challenges
 - Nonlinear stabilization



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Adaptive Mesh Refinement

Convection-Diffusion-Reaction Equations

- **Motivation**
 - Uniform Mesh Refinement becomes **Computationally Expensive**
 - Fine Meshes are only needed in **Localized Regions**



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- **Key Idea**
 - Estimate local discretization errors using **A Posteriori Error Estimators**
 - Error bounded by the estimator

$$\|u - u_h\| \leq C \sum_{K \in \mathcal{T}_h} \eta_K$$



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- Large values of η_K indicate regions **Requiring Refinement**
- **Adapt** the mesh **Automatically** to improve accuracy efficiently



Residual-Based Error Estimators

Convection-Diffusion-Reaction Equations

- **Key Idea**¹
 - Measure how well the numerical solution satisfies the PDE locally
 - **Large Residual** $\Rightarrow$ **Large Local Discretization Error**

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- Typical estimator:

$$\eta_K^2 = h_K^2 \|R_K\|_{L^2(K)}^2 + h_E \|J_E\|_{L^2(E)}^2$$

where J_E denotes edge jump terms

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- **Advantages**

- **Fully Computable** and **Inexpensive**
- **Detects Layers** and **Singularities** effectively

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- **Key Idea**¹
 - Recover an **Improved Gradient** from the computed solution
 - Compare **Recovered Gradient** with **FEM Gradient**

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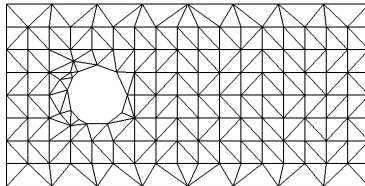
- **Limitations**

- **Less Reliable** for strongly convection-dominated problems
- **Difficulties** near Discontinuities and Sharp Layers

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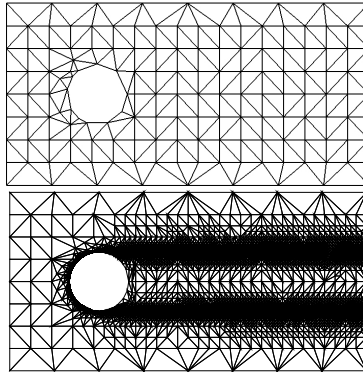


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- **Summary**

- **Finite Element Methods** provide a **flexible framework** for solving PDEs
- **Convection-Dominated** problems require specialized stabilization techniques
- **Adaptive Mesh Refinement** improves efficiency and accuracy
- Modern FEM combines:
 - **Mathematical Analysis**
 - **Scientific Computing**
 - **High-performance Simulations**
 - **Data-Driven Methodologies**



- Modern Research Directions
 - Time-dependent problems
 - Parabolic PDEs

$$\partial_t u - \epsilon \Delta u + \mathbf{b} \cdot \nabla u + cu = f \quad \text{in } \Omega \times [0, T]$$

- Time-Fractional Equations

$$\partial_t^\alpha u - \epsilon \Delta u + \mathbf{b} \cdot \nabla u + cu = f \quad \text{in } \Omega \times [0, T]$$

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- Virtual Element Methods (VEM)¹
- Physics-Informed Neural Networks (PINNs)²

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“Modern FEM research is increasingly moving toward Robust, Adaptive, and Scalable Methods for Multiscale and Multiphysics Systems”

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Thank You!

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